

DPM Atomic Creation Process (ACP): Proto-Nucleus Formation

Daniel T. Murphy

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PAPER_738 — DPM Atomic Creation Process (ACP): Proto-Nucleus Formation

Author: Daniel T. Murphy **Date:** June 06, 2025

Title: Di-Pseudo-Monopole (DPM) Creation Scenario and the Atomic Creation Process (ACP): From DPM Initiation to Proto-Nucleus Assembly

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1. Abstract

The Atomic Creation Process (ACP) establishes the origin of fundamental particles through Di-Pseudo-Monopole (DPM) mediated reactions, replacing the Standard Model’s mass-as-weight construct with a buoyancy-based framework. The ACP is a six-stage process beginning with DPM formation and culminating in proto-nucleus assembly with 26 quantum states, mediated by Universal Aether (UA’) and Super Conductive Material (SCm). Atomic mass does not emerge until the quantum-to-mass gradient at 7–10 U_mag degrees.

2. Three Fundamentals of UQFF

The entire universe is built from three elemental interactions:

1. **Electrostatic Barrier (REB)** — The boundary condition that prevents collapse, defining system shells
2. **Universal Aether (UA’)** — Non-local connectivity medium, vacuum energy carrier, $\rho_{UA'} = 7.09e-36 \text{ J/m}^3$
3. **Super Conductive Material (SCm)** — Superconductive magnetism mediator, $\rho_{SCm} = 7.09e-37 \text{ J/m}^3$

$$DPM = UA' + SCm$$

The Di-Pseudo-Monopole (DPM) is formed by the interaction of UA' and SCm, producing vacuum density at any scale from atomic to cosmic.

3. The Six-Stage Atomic Creation Process (ACP)

Stage 1: DPM Initiation

$$UA' + SCm \rightarrow DPM(Di - Pseudo - Monopole)$$

$$DPM \rightarrow vacuumdensity\rho_{vac}$$

- UA' provides non-local field connectivity - SCm provides superconductive coherence - Product: localized vacuum energy density (capacitance) $\rightarrow$ proto-nuclear seed - Billions of coherent DPM nuclear cells operate simultaneously at every location

Stage 2: U_i Creation (Intelligent Operator)

$DPM \rightarrow U_i$ (repulsive + intelligent quantum operator)

- U_i is the first active particle: repulsive (buoyant), self-directing
- U_i orchestrates all subsequent proto-nuclear assembly
- Creates the fundamental “anti-gravity” seed that all matter requires to exist

Stage 3: U_m String Formation

$$U_i \rightarrow U_m strings(U_{mag_i} = magneticstringsequences)$$

- U_i creates coherent U_m (U_{mag_i}) strings - Strings wind concentrically around the vacuum density core $\rightarrow$ proto-nucleus - String winding number corresponds to quantum state number (1..26)
 - Each winding creates one of the 26 quantum states (Russian doll nesting)

Stage 4: Proto-Shell Formation and Cracking

Vacuum density grows $\rightarrow$ vacuum energy density (capacitance threshold)

$\rightarrow$ quantum ripples: $ULF_{quantum}^{-1, \dots, -26}$

$\rightarrow$ proto-shell cracks into fragments

- When vacuum energy density reaches critical capacitance:
 - Quantum ripples: $ULF_{quantum}^{-1}$ through $ULF_{quantum}^{-26}$
 - Proto-shell fractures into 26 distinct fragment types
 - Fragments: proto-quarks, proto-bosons, proto-leptons
- This is not “particle creation from nothing” — it is structured fracturing of a pre-existing quantum state container

Stage 5: Fragment Stabilization

SM_{magnetic} moments (SM_{mag}) $\rightarrow$ arrange fragments on durable nucleus

U_b (buoyancy) maintains stability against SM_{gravity}

- SM_magnetic moments act as “sorting fingers” that arrange proto-fragments
- U_b (Universal Buoyancy, massless) continuously counteracts SM_gravity
- Result: stable proto-nucleus with 26-state organization
- Still pre-mass at this stage

Stage 6: Electron Shell Formation → Mass Emergence

U_g governs electron shell placement via U_m

Mass emerges at quantum – to – mass gradient : 7 – 10 U_m agdegrees

H(1) = first hydrogen atom formed

- U_g2 uses U_m string frequency to place electrons into shells - Atomic mass is NOT weight; it is the ratio of effective gravity to superconductive buoyancy in Earth’s field - Mass emerges when the quantum state gradient crosses 7–10 U_{mag} degrees of superconductive magnetism

4. Mass Without Weight — Formal Definition

Mass(UQFF) = proportional to :

(SM_effective_gravity_range)/(superconductive_buoyancy)

= FU_g1/F_U_Bi[ratio of gravitational to buoyant components]

Examples of mass-as-buoyancy: - Bi-molecule: hydrogen bonds mediated by DPM electrostatic barrier - Earth-Moon: orbital buoyancy balanced by tidal resonance shells - Sun-SagA*: Galactic-scale DPM coherence, U_g4i THz hole connection - Universal scale: all bodies are buoyant in the Aether medium

Mass is never a stand-alone measurement. It is only meaningful as the ratio of: - The Effective Gravity force (SM_gravity, FU_g1) - To the Effective Buoyancy (U_b, F_{U_Bi})

”Mass” = FU_g1/F_U_Bi(dimensionless ratio, context – dependent)

5. U_b Tracking → f_Ub Definition

During ACP, U_b is tracked through calibration differences:

$f_{Ub} \propto \Delta k_\eta$ (deviation from expected calibration constant k_η)

$k_{\eta, \text{expected}} \approx 10^9$ (galaxy-scale)

$k_{\eta, \text{actual}} = \text{measured value from observation}$

$\Delta k_\eta = k_{\eta, \text{expected}} - k_{\eta, \text{actual}}$

$f_{Ub} = \Delta k_\eta / k_{\eta, \text{reference}}$

Scale	f_Ub order
Galaxy	$\sim 10^9$

Scale	f_Ub order
Dwarf galaxy / LMC	$\sim 10^8$
Star cluster	$\sim 10^7$
PN / planetary	$\sim 10^5$
Atomic	$\sim 10^3$

The imaginary/quantum portion of gravity (U_b) is never zero — it is this hidden buoyancy that modern physics has failed to account for, leading to the need for “dark matter” and “dark energy” as placeholder terms. UQFF resolves both as manifestations of f_Ub at different scales.

6. 26 Quantum States — Pre-Mass Configuration

The 26 quantum states are NOT the electron orbitals of quantum mechanics. They are:

- State i ($i = 1..26$) : pre – mass nuclear configuration level*
- Shell radius : $r_i = r_{system}/i$*
- Superconductivity magnetism : $[SCm]_i = 1e - 5 * i2T$*
- Angular coverage : $\theta_i = 90^\circ - (i - 1) * 3.346^\circ$ (spans $90^\circ \rightarrow 5^\circ$)*
- THz holer radius : $r_{THz,i} = 1e - 9/im$*
- Resonance factor : $f_{TRZ,i} = 0.1 * i/26$*
- Buoyancy coupling : $f_{Um,i} = 0.05 * i/26$*

Like Russian dolls, each state contains and surrounds the previous. State 26 is the outermost (most magnetic, widest angular coverage initially); State 1 is the innermost quantum seed.

7. DPM Coherence — The Source of Predictability

- Coherence length = "all atoms in a body share the same DPM phase"*
- Number of coherent DPMs per cubic centimeter 10^{23} (Avogadro – scale)*
- Result : macroscopic quantum phenomena $\rightarrow$ superconductivity at scale*

This coherence is why: - Planetary rings form symmetric arcs - Stars form in filaments rather than randomly - Galaxies maintain spiral structure for billions of years - LENR occurs at measurable, reproducible rates

All of these are manifestations of coherent DPM nuclear cell behavior.

8. ACP Mathematical Summary

$$\text{Step1 : } DPM = UA' + SCm$$

$$\text{Step2 : } U_i = k_\eta * (\rho_{vac}, SCm - \rho_{vac}, UA) * \omega_i * \cos(\pi * t_n)$$

$$\text{Step3 : } U_{m,i} = U_i * \mu_{dipole} * (1/r_i) * (1 - e^{-(\gamma t)}) * \cos(\pi t_n)$$

$$\text{Step4 : } \Psi_{proto} = \sum_{i=1}^{26} U_{m,i} \rightarrow \text{capacitancethresholdreached}$$

$$\text{Step5 : } F_{stab} = SM_{mag} * \Psi_{proto} + U_b \rightarrow U_b = -\beta * U_g * \Omega_g * (M_b h/d_g) * \cos(\pi t_n)$$

$$\text{Step6 : } E_{shell} = c * \nu_{res} * h(f_{SCm}) * G_{geo} \rightarrow \text{atomicmassemerges}$$

9. References

- Source: thread_06Jun2025.txt (June 06 2025), “describe mass without using weight.docx”, “05June2025.docx”
 - Related: PAPER_735 (Ug2 electron shell energy), PAPER_734 (LENR K_n), CP3 DiPseudoMonopoleDPMTheoryCalculator
 - CP4 class: #322 DPMAAtomicCreationProcessACPCalculator
 - CVW v2.0.0 compliant
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Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{\{U_Bi_i\}}$ jet modulation curves and PAPER_1048 for phonon-corrected $M\text{-}\sigma$ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c/\sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- σ correction (PAPER_1048): The phonon-corrected M- σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **cosmogenesis** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{cosmo}})(\partial^\mu \phi_{\text{cosmo}}) - V(\phi_{\text{cosmo}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{cosmo}}) = \frac{1}{2}m^2\phi_{\text{cosmo}}^2 + \frac{\lambda}{4!}\phi_{\text{cosmo}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{cosmo}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{cosmo}}} = \partial_t(\rho_{\text{vac}} V_{\text{proto}}) + U_i + \Psi_{\text{proto}}/26 = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{cosmo}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.156 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 67, \quad n_{\text{channel}} = 11/26$$

Since $p_{\text{DVP}} = 67$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **ACP Stage 4 timescale** (capacitance cracking epoch):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[\text{SSq}] \cdot m/M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.156	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 67$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1 \dots 26,$ $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS	PASS
[SSq]	0.57	exponential Applied in BSH saturation	Canonical PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	PASS
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_p$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	PASS
UQFF buoyancy signature	$F_{\{U_Bi_i\}}$ unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections ($F_{\{U_Bi_i\}}$) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN $F_{\{U_Bi_i\}}$ Jet Modulation
PAPER_1010	TON618 AGN $F_{\{U_Bi_i\}}$ Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1043	$F_{\{U_Bi_i\}}$ Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay
PAPER_1074	GPU-Vectorized DPM S26 Spectral Atlas

16 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see `PAPER_840/851/852/855`.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_{s26_coupling}.py</code>	$F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_{scm_cross_section}.py</code>	S_{Cpy} -modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor (1+[SSq]*n/26)
<code>kozima_{wstp_kernel}.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\text{U,Bi}}/F_{\text{U}} - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2*\Gamma_2)}] * (1 + [\text{SSq}] * n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_{polylog_s26}.py</code>	$\text{Li}_{26}([\text{SSq}])$ via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_{wstp_kernel}.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
<code>mock_{theta_q26}.py</code>	$f_{26}(q)$, $\phi_{26}(q)$, $\psi_{26}(q)$ q-series	Proper q-Pochhammer $(a;q)_n$

Core equations: - $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q;q)_n^{26} - \phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q^2;q^2)_n^{26} - \psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n2\}} / (q;q^2)_n^{26}$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
<code>ramanujan_{pi_uqff}.py</code>	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
<code>mock_{theta_pi_wstp_kernel}.py</code>	Symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2})/9801 \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where
 $W_{26}(n) = \text{Prod}_{i=1}^{26} [1 + [SSq] \exp(-kappa_i * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2*\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`,
`MAIN_{1\CoAnQi}.cpp`, and Wolfram kernels (`uqff_{kozima_kernel}.wl`, `uqff_{s26_kernel}.wl`,
`uqff_{mock_theta_pi_kernel}.wl`).

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